

Team Paradox — SIH 2026 Complete Research & PS Recommendation Report

Team: Team Paradox | **Institute:** IET Gorakhpur (DDU University)

Date: August 27, 2026 | **Report Version:** 2.0 (Consolidated)

Scope: 226 SIH 2026 PS analyzed, 18 PS deep-researched, team skill-mapped, final recommendations delivered

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1. Executive Summary

This report consolidates **three independent research streams** into one master document:

Stream	Source	PS Covered
Analyst Research	Deep analysis of 6 organizations (41 PS) + GitHub profile analysis	SIH26047, SIH26018, SIH26111, SIH26127 + 3 hardware PS
Abhiuday's Picks	7 PS across disaster, cybersecurity, healthcare, monitoring	SIH26008, SIH26068, SIH26095, SIH26106, SIH26133, SIH26152, SIH26208
Om's Picks	2 PS focused on land governance & GIS	SIH26013, SIH26014
Arun's Picks	2 PS in disaster management & cybersecurity	SIH26001, SIH26106

Final Verdict: After analyzing all 18 unique PS against team skills, existing projects, competition levels, and winning probability, **7 PS emerge as viable options** ranked from best-fit to stretch goals.

2. Team Skill Assessment

2.1 Member Profiles

Member	Role	Key Strength	Skill Level
Aditya Chaudhari	Team Lead · Backend	Python/Flask, ESP32/IoT, WebSocket, OCR (Azure), Selenium, 8+ languages	★★★★★
Anshika Singh	Frontend · CV · Security	React/Flutter, Crop Disease CNN, Face Detection, Phishing Detection Ext, SIH 2025 exp	★★★★☆
Om Abhishek Tripathi	AI/ML · Full-Stack	ARIA AI assistant, Track-My-Bus, ArduPilot (drones), ECWoC/AISSOCE open-source	★★★★☆
Abhiuday Pratap Singh	Frontend · DSA	TypeScript/React, Sorting Visualizer, LeetCode, Hackathon exp (ignite-26, RIFT-26)	★★★★☆
Anoop Shukla	Web Dev · Support	190 contributions/yr, Campus Connect project, GitHub Pro	★★★☆☆
Arun Kumar	AI/ML · Research	CSE AI/ML academic background (minimal GitHub — unproven coding)	★★★☆☆

2.2 Team Skill Matrix

Skill Area	Aditya	Anshika	Om	Abhiuday	Anoop	Arun	Team Total
Python Backend	★★★★	★★★	★★★	★★★	★★	★★	Strong
Frontend (React/TS)	★★★★	★★★★★	★★★★★	★★★★★	★★★★	★	Strong
Mobile (Flutter)	★★★★★	★★★	★★★	★	★	★	Strong
Computer Vision	★★★★	★★★	★★	★	★	★	Moderate
ML/AI Models	★★	★★	★★★	★	★	★★	Moderate
NLP/Text Processing	★★	★	★★	★	★	★	Weak
Database Design	★★★★	★★	★★	★★	★★	★	Moderate
DevOps/Cloud	★★	★★	★★★	★	★	★	Weak
Cybersecurity	★★	★★★★	★★	★	★	★	Moderate
IoT/Hardware	★★★★	★	★★	★	★	★	Weak
GIS/Geospatial	★	★	★	★	★	★	NONE
DSA/Algorithms	★★★★	★★★★	★★★★	★★★★★	★★	★★	Strong

2.3 Team Strengths & Gaps

✓ Strengths:

- Full-stack web (Python Flask + React/TS + Flutter)
- Real-time systems (WebSocket LAN Messaging)
- Basic CV (face detection, crop disease, deep-face)
- OCR (Aadhar Scanner with Azure Vision API)
- Automation (Selenium web scraping)
- SIH experience (BlueAlert_SIH2025)
- Conversational AI (Om's ARIA assistant)
- Phishing detection (Anshika's browser extension)

✗ Critical Gaps:

- No production ML model training experience
- Zero GIS/geospatial capability
- No deep cybersecurity (network forensics, cryptography)
- No hardware prototyping beyond basic ESP32
- No domain expertise (AYUSH, dairy, railways, land governance)

- No large-scale distributed systems experience

3. SIH 2026 Landscape Overview

Key Statistics

- **226 Total PS** (172 Software + 54 Hardware)
- **30 Organizations** (27+ ministries, 2 state govs, 12 PSUs/corporates)
- **18 Themes** across domains
- **50,000+ teams** expected to participate
- **₹1,00,000 prize** per PS winner
- **Idea Submission Deadline:** September 20, 2026
- **Grand Finale:** December 2026 (36-hour non-stop hackathon)

Difficulty Distribution (SIH 2026)

Level	Score	Characteristics	PS Count	Share
Level 1 — Easy	0-39	CRUD, public datasets, rapid prototyping	78	34.5%
Level 2 — Medium	40-59	Multi-system integration, applied ML/NLP/CV	66	29.2%
Level 3 — Hard	60-79	Specialized algorithms, custom PCB, crypto	45	19.9%
Level 4 — Very Hard	80-100	Mission-critical, defense, deep HW-SW co-design	37	16.4%

What Wins SIH (Based on 2020-2025 Winners Analysis)

1. **Working prototype** (not just slides/mockups)
2. **One core feature done perfectly** (not 10 features half-done)
3. **Integration with existing government systems** (ABDM, Bhuvan, NTES, DILRMP)
4. **Domain understanding** (judges ask "Have you talked to actual users?")
5. **Scalability plan** (how will this work at national scale?)
6. **Cost analysis** (government solutions must be affordable)

4. Primary Recommendations (Analyst's Pick)

Based on deep analysis of 41 PS from 6 key organizations + team GitHub profile analysis

#1: SIH26047 — AYUSH Patient Case-Taking Software

Attribute	Detail
Organization	Ministry of AYUSH
Category	Software
Difficulty	Medium (6/10)
Expected Crowd	200-300 teams
Team Fit Score	★★★★★ (95%) — HIGHEST FIT
Winning Probability	HIGH

Why: Team has **4 directly overlapping projects**: Edu-Portal (management system), deep-face-python (facial analysis → prakriti assessment), crop-disease-detection (image classification → symptom classification), Aadhar Scanner (OCR → medical record digitization).

What to Build: Digital case-taking platform for AYUSH practitioners supporting Ayurveda, Unani, Siddha, Homoeopathy with prakriti assessment, Nadi Pariksha recording, classical text reference, treatment planning, and ABDM integration.

Tech Stack: Flask/FastAPI + MongoDB + React.js + Flutter + NLP (spaCy) + ABDM APIs

Role Assignments:

Member	Role	Reason
Aditya (Lead)	Backend + API Design + Auth	Strongest Python backend, Edu-Portal experience
Anshika	Frontend Dashboard + Flutter Mobile	Strongest frontend/Flutter, medical classification UI
Om	AI/ML Module + NLP	Strongest AI interest, ARIA AI assistant experience
Abhiuday	Frontend Components + Charts	TypeScript/React proficiency
Anoop	Documentation + API Testing + UI	Supporting tasks
Arun	AYUSH Research + Data Compilation	Academic research for domain knowledge

#2: SIH26111 — AI Feed & Silage Quality Testing

Attribute	Detail
Organization	Ministry of Fisheries, AH & Dairying
Category	Software
Difficulty	Medium (6/10)
Expected Crowd	150-200 teams
Team Fit Score	★★★★☆ (85%)
Winning Probability	MEDIUM-HIGH

Why: Anshika's crop-disease-detection project is **DIRECTLY applicable** — same CNN architecture, different dataset (feed quality vs crop disease). NIR spectroscopy simulation with published datasets.

#3: SIH26018 — Intelligent Land Record Digitization & Validation

Attribute	Detail
Organization	Ministry of Rural Development
Category	Software
Difficulty	Medium-Hard (7/10)
Expected Crowd	120-180 teams
Team Fit Score	★★★★☆ (80%)
Winning Probability	MEDIUM-HIGH

Why: Aditya's Aadhar Scanner (OCR with Azure Vision API) is **directly applicable** to land record digitization. OCR + NLP extraction + validation = your team has built this before.

#4: SIH26127 — City-Wide ANPR Traffic Analytics

Attribute	Detail
Organization	MoHUA
Category	Software
Difficulty	Hard (8/10)
Expected Crowd	150-200 teams
Team Fit Score	★★★☆☆ (70%)
Winning Probability	MEDIUM — Ambitious

Why: Team's CV skills (face detection, crop disease) + Om's Track-My-Bus + WebSocket real-time = foundation exists. But ANPR is technically demanding.

5. Team Members' PS Suggestions — Deep Analysis

Summary of Team Members' Suggestions

Member	PS Suggested	Count
Abhiuday Pratap Singh	SIH26008, SIH26068, SIH26095, SIH26106, SIH26133, SIH26152, SIH26208	7
Om Abhishek Tripathi	SIH26013, SIH26014	2
Arun Kumar	SIH26001, SIH26106	2
Total Unique PS		10

Note: SIH26106 was suggested by both Abhiuday and Arun — strong team interest signal.

5.1 SIH26068 — WeatherGPT: Conversational AI for Weather Forecasting

Suggested by: Abhiuday

Attribute	Detail
Organization	Ministry of Earth Sciences (IMD)
Category	Software
Theme	Disaster Management
Difficulty	Medium (6/10)
Expected Crowd	150-250 teams
Team Fit Score	★★★★☆ (80%)
Winning Probability	HIGH

What It Is: Build an AI-powered chatbot ("WeatherGPT") that integrates meteorological datasets, forecasting models, and disaster warning systems to provide accurate, contextual, multilingual weather intelligence through conversational interfaces.

Key Features Required:

- Real-time weather data retrieval from IMD APIs
- NLP-based natural language querying for forecasts
- Integration with NWP models (GFS/WRF)
- Extreme weather alerts and early warnings
- Location-based forecasting and advisory
- **Multilingual support** (Indian languages)
- Voice-enabled interaction for rural accessibility
- Climate trend and historical analysis

Suggested Tech Stack (from PS): Python/FastAPI/Node.js, MQTT/WIS2.0/WebSocket, LLMs (OpenAI, Llama, Gemini), GIS tools, PostgreSQL/MongoDB, Docker/Kubernetes

 Why This Is GREAT for Team Paradox:

Team Project	Direct Overlap
Om's ARIA AI Assistant	Conversational AI architecture — same pattern, different domain
Aditya's LAN Messaging	WebSocket real-time data streaming
Team's Flask/FastAPI skills	Backend API development
Team's Hindi proficiency	Multilingual support (key requirement)
Aditya's Selenium automation	Data scraping from IMD portals
Anshika's Flutter skills	Mobile app development

What You Need to Learn:

- IMD weather APIs (mausam.imd.gov.in, OpenWeatherMap, Visual Crossing)
- LLM integration (OpenAI/Gemini API with weather-specific prompting)
- Basic GIS for location-based queries (Mapbox or Leaflet)
- Voice interaction (Web Speech API or Whisper for Hindi/regional languages)

36-Hour Feasibility: ★★☆☆☆ — Core chatbot + weather data + alerts is achievable. Voice + multilingual is stretch.

Competitive Analysis:

- IMD PS attract diverse teams (not just CS — also meteorology students)
- ChatGPT wrappers are common — you need deep IMD integration to stand out
- Key differentiator: offline functionality + voice in Hindi/Bhojpuri for UP farmers

★ **VERDICT: STRONG RECOMMENDATION — Add to Top 5**

Role Assignments:

Member	Role
Aditya	Backend (FastAPI + IMD API integration + caching + WebSocket alerts)
Om	LLM integration (prompt engineering + RAG for weather data + ARIA architecture transfer)
Anshika	Flutter mobile app + voice interaction + map visualization
Abhiuday	React web dashboard + weather visualization (D3.js charts)
Anoop	API testing + documentation + alert notification system
Arun	IMD data research + agricultural use-case compilation + test dataset

5.2 SIH26095 — Smart Real-Time Monitoring & Inspection Mobile App

Suggested by: Abhiuday

Attribute	Detail
Organization	Ministry of Social Justice and Empowerment (MoSJE)
Category	Software
Theme	Miscellaneous
Difficulty	Medium (5/10)
Expected Crowd	100-180 teams
Team Fit Score	★★★★☆ (82%)
Winning Probability	HIGH

What It Is: A centralized mobile app for real-time monitoring, surprise inspections, CCTV surveillance integration, and random inspection assignment for projects/institutes/NGOs running under DoSJE schemes.

Key Features Required:

- Live CCTV feed integration from projects/institutes
- Random Video Conferencing (VC) with Project Incharge/Staff/Beneficiaries
- Real-time monitoring dashboard for Department officials
- Mobile-based inspection module for PMU/Inspection Teams
- **Random assignment of inspection duties through AI/automation**
- Geo-tagged inspection reports and live evidence capture
- AI-based anomaly and attendance analytics

🎯 Why This Is GREAT for Team Paradox:

Team Project	Direct Overlap
Aditya's Edu-Portal	Management system architecture (student management → NGO/project management)
Aditya's LAN Messaging	Real-time WebSocket communication (CCTV streams, VC)
Anshika's Flutter skills	Mobile inspection app
Aditya's Captive Portal	Network/security understanding for CCTV integration
Team's React dashboard	Real-time monitoring dashboard
Aditya's Selenium	Automated monitoring/testing patterns

What You Need to Learn:

- RTSP/HLS streaming for CCTV integration (WebRTC or similar)

- Geo-tagging (Google Maps API / Mapbox)
- Random assignment algorithm (weighted random + constraint-based)
- Basic anomaly detection (attendance analytics)

36-Hour Feasibility: ★★★★★ — This is essentially a CRUD + dashboard + mobile app + CCTV viewer. Very achievable.

Competitive Analysis:

- "Miscellaneous" theme = less niche competition
- MoSJE PS attract fewer elite teams (not as trendy as cybersecurity or AI)
- Key differentiator: working CCTV integration + real AI anomaly detection
- Lower competition than AYUSH or Railway PS

★ **VERDICT: STRONG RECOMMENDATION** — Easiest to execute, very achievable

Role Assignments:

Member	Role
Aditya (Lead)	Backend (Flask + WebSocket + CCTV RTSP integration + random assignment engine)
Anshika	Flutter mobile inspection app + geo-tagged reports + evidence capture
Om	AI anomaly detection (attendance patterns + behavior analytics) + random VC connector
Abhiuday	React real-time dashboard + monitoring views + alert system
Anoop	API testing + documentation + notification system
Arun	DoSJE scheme research + NGO/institute data compilation + compliance rules

5.3 SIH26133 — Accessibility & Quality of Public Healthcare in Rural Areas

Suggested by: Abhiuday

Attribute	Detail
Organization	Government of Maharashtra (State Innovation Society)
Category	Software
Theme	MedTech / HealthTech
Difficulty	Medium (6/10)
Expected Crowd	200-300 teams
Team Fit Score	★★★★☆ (85%)
Winning Probability	HIGH

What It Is: An integrated care-access and quality support solution combining assisted teleconsultation, appointment/queue management, digital triage, longitudinal patient records, referral tracking, diagnostic coordination, medicine availability, high-risk patient follow-up, and facility dashboards.

Key Features Required:

- Assisted teleconsultation for rural patients
- Appointment and queue management
- Digital triage system
- Longitudinal patient records (interoperable, ABDM standards)
- Referral tracking (sub-centre → PHC → rural hospital → district hospital)
- Diagnostic coordination and medicine availability tracking
- High-risk patient follow-up alerts
- Facility dashboards
- Multilingual, low-connectivity/offline support
- Frontline health worker support

🎯 Why This Is GREAT for Team Paradox:

Team Project	Direct Overlap
Aditya's Edu-Portal	Management system → healthcare facility management
Aditya's Aadhar Scanner	Patient identification + record linking
Anshika's Flutter	Mobile app for frontline health workers
Team's React dashboard	Facility dashboards
Aditya's WebSocket	Real-time queue management
Team's SIH26047 research	AYUSH case-taking overlaps heavily with rural healthcare

What You Need to Learn:

- ABDM (Ayushman Bharat Digital Mission) APIs
- Healthcare data standards (HL7 FHIR)
- Digital triage algorithms (symptom-based priority scoring)
- Offline-first architecture (PWA with service workers)

36-Hour Feasibility: ★★☆☆☆ — Core features very achievable. ABDM integration is stretch.

Competitive Analysis:

- Healthcare PS always attract many teams (200-300)
- But this is a STATE GOVERNMENT PS (Maharashtra) — judges may be state officials who value practical, deployable solutions over flashy tech
- Key differentiator: offline-first + multilingual (Marathi/Hindi) + actual health worker UX
- Very similar to SIH26047 (AYUSH case-taking) — **team can prepare for both simultaneously!**

★ **VERDICT: EXCELLENT PICK — Very similar to #1 recommendation (SIH26047). Can prepare for both!**

Role Assignments:

Member	Role
Aditya (Lead)	Backend (Flask + patient records + referral engine + ABDM integration)
Anshika	Flutter health worker app + teleconsultation UI + offline PWA
Om	Digital triage AI + high-risk patient prediction + appointment optimization
Abhiuday	React facility dashboard + queue management + real-time stats
Anoop	Testing + documentation + medicine availability module
Arun	Maharashtra healthcare data research + PHC/sub-centre structure + ABDM standards

5.4 SIH26106 — AI-Powered Email Threat Detection & Forensic Intelligence

Suggested by: Abhiuday **AND** Arun (double-nominated — strong team interest!)

Attribute	Detail
Organization	AICTE (Cyber Security Cell)
Category	Software
Theme	Blockchain & Cybersecurity
Difficulty	Hard (8/10)
Expected Crowd	200-350 teams (cybersecurity is trendy)
Team Fit Score	★★★☆☆ (65%)
Winning Probability	MEDIUM — Ambitious

What It Is: An AI-powered platform that detects phishing/spoofed/fraudulent emails in real-time, analyzes complete email technical structure (headers, SMTP relay paths, SPF/DKIM/DMARC), traces transmission path across mail servers, estimates origin with geolocation, and generates forensic intelligence for investigators.

Key Components Required:

1. **Fraud Detection Engine:** NLP for urgency cues, impersonation language, social engineering
2. **Header Analysis Module:** Deep SPF/DKIM/DMARC validation, relay path analysis
3. **Origin Traceability:** IP geolocation, WHOIS/DNS analysis, VPN/TOR detection
4. **Identity Correlation:** Graph-based relationship analysis, threat intelligence correlation
5. **Dashboard & Reporting:** Fraud score, trace maps, forensic reports, case management
6. **Privacy & Compliance:** Evidence preservation, chain-of-custody, data masking

🎯 Team Fit Analysis:

Team Project	Overlap
Anshika's Phishing Detection Extension	✅ Direct — browser-based phishing detection. But PS requires server-side deep analysis
Aditya's Captive Portal	Partial — networking understanding
Aditya's Python automation	✅ Scraping/data pipeline skills
Om's DebateAI	Partial — NLP understanding

What You Need to Learn (a LOT):

- SMTP protocol internals (Received headers, relay chains, envelope vs header)
- SPF/DKIM/DMARC validation (DNS TXT records, cryptographic signatures)
- IP geolocation APIs (MaxMind, IPinfo)
- WHOIS/DNS forensics
- Graph databases (Neo4j) for relationship mapping

- NLP for social engineering detection (urgency, authority, scarcity cues)

36-Hour Feasibility: ★★☆☆☆ — Tight. Core detection + header analysis is doable. Full forensic tracing is stretch.

Competitive Analysis:

- Cybersecurity PS attract specialized teams (many have CTF experience)
- AICTE PS = high visibility, judges may be cybersecurity professionals
- Key differentiator: visual forensic reports + AI-based attribution confidence scoring
- Anshika's phishing detection extension is a genuine head-start that most teams won't have

⚠ VERDICT: AMBITIOUS — Good pick if team is passionate about cybersecurity. Requires 3-4 weeks of learning email forensics before SIH.

Role Assignments:

Member	Role
Aditya (Lead)	Backend (FastAPI + email parsing + header analysis + relay path reconstruction)
Anshika	Phishing detection NLP + dashboard UI + browser extension integration (extend existing project!)
Om	ML classification (legitimate vs phishing vs BEC) + graph analysis (Neo4j) + threat intelligence
Abhiuday	React forensic dashboard + trace visualization + map-based geolocation display
Anoop	API testing + documentation + case management module
Arun	Email protocol research + threat intelligence compilation + test email dataset creation

5.5 SIH26152 — Social Media Analytics for Intelligence

Suggested by: Abhiuday

Attribute	Detail
Organization	NTRO (National Technical Research Organisation)
Category	Software
Theme	Miscellaneous
Difficulty	Hard (7/10)
Expected Crowd	150-250 teams
Team Fit Score	★★★☆☆ (55%)
Winning Probability	MEDIUM-LOW

What It Is: AI-driven Social Media Analytics Framework that processes raw platform data to extract: sentiment analysis, demographic profiling, trend detection, and link/network analysis. Must support X (Twitter), Telegram (essential), Instagram/Facebook (desirable), Reddit/YouTube (appreciable).

5 Core Components:

- A. Multi-platform data ingestion pipeline (X, Telegram, Instagram, etc.)
- B. Multi-dimensional sentiment inference (sarcasm, anxiety, excitement detection)
- C. Automated demographic profiling (age, geography, language, interests)
- D. Real-time trend & topic detection
- E. Link analysis & network topology (influence mapping)

🎯 Team Fit Analysis:

Requirement	Team Capability	Gap
Data ingestion pipeline	✅ Aditya's Selenium + Python automation	Can build scrapers
Sentiment analysis (NLP)	⚠️ Basic NLP only	Need to learn transformer models
Demographic profiling	❌ No experience	Hard ML problem
Trend detection	⚠️ Can use existing libraries	Achievable with LDA/BERTopic
Network/Link analysis	❌ No graph analytics experience	Need NetworkX/Neo4j
Multi-platform APIs	⚠️ Basic API skills	X API v2 has strict rate limits

36-Hour Feasibility: ★★★☆☆ — Very tight. Building all 5 components in 36 hours is extremely ambitious.

⚠️ **VERDICT: RISKY** — Impressive PS but team lacks NLP depth and graph analytics. Only attempt if willing to use pre-built models heavily.

5.6 SIH26208 — Student Innovation (Toys & Games — Heritage & Culture)

Suggested by: Abhiuday

Attribute	Detail
Organization	AICTE (MIC-Student Innovation)
Category	Software
Theme	Heritage & Culture
Difficulty	Unknown (vague PS)
Expected Crowd	200-400 teams (open-ended = popular)
Team Fit Score	★ ★ ☆ ☆ ☆ (35%)
Winning Probability	LOW — Too vague to prepare

What It Is: "Challenge your creative mind to conceptualize and develop unique toys and games based on our civilization, history, and culture."

⚠ VERDICT: AVOID — This is an open-ended creative PS with no clear technical requirements. Cannot prepare systematically. Judging criteria are subjective. Team's strength is in structured engineering, not creative game design. Skip.

5.7 SIH26008 — Conveyor Belt Monitoring in Iron Ore Mining (HARDWARE)

Suggested by: Abhiuday

Attribute	Detail
Organization	Ministry of Steel (NMDC)
Category	HARDWARE
Theme	Smart Automation
Difficulty	Hard (8/10)
Expected Crowd	20-40 teams
Team Fit Score	★ ★ ☆ ☆ ☆ (25%)
Winning Probability	LOW — Heavy industrial IoT, mining domain

What It Is: Intelligent Conveyor Belt Health Monitoring using IoT sensors (vibration, temperature, acoustic, load), AI vision monitoring (cracks/tears via cameras), drone inspection, digital twin, SCADA/PLC integration, and ML-based predictive analytics for iron ore mining.

⚠ VERDICT: AVOID — Requires industrial IoT expertise, mining domain knowledge, SCADA/PLC integration, vibration/acoustic sensor calibration. Far beyond team's hardware capability. ESP32 WiFi extender ≠ industrial mining IoT.

5.8 SIH26013 — Geospatial Data Harmonization for Land Records

Suggested by: Om

Attribute	Detail
Organization	Ministry of Rural Development (DoLR)
Category	Software
Theme	Disaster Management (Land)
Difficulty	Hard (8/10)
Expected Crowd	80-120 teams
Team Fit Score	★ ★ ☆ ☆ ☆ (30%)
Winning Probability	LOW

What It Is: AI-enabled geospatial integration platform that automatically integrates drone imagery, orthorectified imagery, DSM/DTM, cadastral maps, revenue records, municipal GIS layers, utility networks, ground truthing data, GNSS/CORS data, and building footprints. Requires spatial matching, topology correction, coordinate transformation, change detection, and conflict resolution.

Required Tech: AI/ML, GeoAI, GIS & WebGIS, Spatial Databases, ETL Automation, Computer Vision, Cloud Computing, Spatial Analytics

⚠ VERDICT: AVOID — Team has ZERO GIS/geospatial skills. This PS requires PostGIS, GeoServer, QGIS, drone imagery processing, coordinate transformation, and spatial databases. Learning all this from scratch in weeks is not feasible.

5.9 SIH26014 — Integrated GIS-based Land Stack (Digital Public Infrastructure)

Suggested by: Om

Attribute	Detail
Organization	Ministry of Rural Development (DoLR)
Category	Software
Theme	Robotics and Drones
Difficulty	Very Hard (9/10)
Expected Crowd	100-150 teams
Team Fit Score	★☆☆☆☆ (35%)
Winning Probability	LOW

What It Is: Build a prototype of "Land Stack" — an integrated GIS-based Digital Public Infrastructure for Land Governance. Must integrate cadastral maps, Record of Rights, registration records, master plans, building permissions, encumbrances, land use, property taxation, utility infrastructure — all linked via ULPIN (unique land parcel ID). Already piloted in Chandigarh and Tamil Nadu (launched Dec 31, 2025).

⚠ VERDICT: AVOID — Even more GIS-heavy than SIH26013. This is a flagship national-level project that requires deep understanding of Indian land governance, cadastral systems, GIS integration, and state-level data formats. Team has zero GIS capability. However, SIH26018 (Land Record Digitization) from same ministry IS achievable — it focuses on OCR/digitization, not GIS.

5.10 SIH26001 — AI-Based Landslide Early Warning System in NER

Suggested by: Arun

Attribute	Detail
Organization	Ministry of Development of North Eastern Region (MDoNER)
Category	Software
Theme	Disaster Management
Difficulty	Hard (7/10)
Expected Crowd	120-200 teams
Team Fit Score	★★★★☆ (45%)
Winning Probability	MEDIUM-LOW

What It Is: AI-powered early warning and monitoring platform for landslide-prone areas in North Eastern Region. Must collect/analyze rainfall, soil moisture, satellite imagery, terrain/slope data, historical landslide records. Use AI/ML for risk prediction. GIS mapping for visualization. Citizen geo-tagged reporting. Multilingual notifications. Offline functionality.

Team Fit Analysis:

Requirement	Team Capability	Gap
AI/ML prediction	⚠️ Basic — can use pre-trained models	Need landslide-specific features
GIS mapping	❌ Zero GIS skills	Major gap
Satellite imagery analysis	❌ No remote sensing experience	Major gap
Dashboard/web app	✅ Strong React + Flask	Can build
Mobile app (citizen reporting)	✅ Flutter	Can build
Multilingual + offline	⚠️ Basic	Partial
Weather API integration	✅ API skills	Can do

36-Hour Feasibility: ★★☆☆☆ — Dashboard + mobile + basic prediction is doable. GIS + satellite analysis is the bottleneck.

⚠️ **VERDICT: MODERATE** — Interesting disaster management PS, but GIS/satellite requirements are major gaps. Could work if team focuses on the prediction engine + dashboard + citizen reporting (skip deep GIS) and uses pre-built satellite data visualization.

6. Hardware PS Backup Analysis

From the 6 researched organizations (AYUSH, Dairy, Rural Dev, Housing, Railways, NTRO)

HW-1: SIH26048 — iKwath Smart Pod-based Kwatha Maker

Attribute	Detail
Organization	Ministry of AYUSH
Fit	★★★★☆ (55%)
Competition	LOW (30-60 teams)
Core Hardware	ESP32 + PT100 sensor + PID control + NFC + heating element + load cell
Core Software	Embedded C firmware + React Native app + Firebase backend
Aditya's Edge	ESP32 WiFi Extender project → direct microcontroller experience
Verdict	Viable with 4 weeks hardware prep. "Nespresso for Ayurveda" = massive wow factor

HW-2: SIH26109 — AI-Based Bovine Mastitis Prediction

Attribute	Detail
Organization	Ministry of Fisheries, AH & Dairying
Fit	★★★★☆ (60%) — BEST hardware pick
Competition	LOW (40-80 teams)
Core	IoT sensors + ML prediction (XGBoost/TFLite) + LoRa + solar + dashboard
Why Best	Most ML-heavy hardware PS — fits Om's AI trajectory. Low competition.
Verdict	Best hardware option if team invests in sensor learning

HW-3: SIH26110 — Low-Cost Milk Chilling Can

Attribute	Detail
Organization	Ministry of Fisheries, AH & Dairying
Fit	★★★☆☆ (40%) — LOWEST
Competition	VERY LOW (30-50 teams)
Core	Phase Change Materials + thermodynamics + insulation (mostly mechanical)
Verdict	SKIP — Primarily mechanical engineering, team's software skills are irrelevant

7. Master Comparison Matrix — ALL 18 PS

Software PS (Ranked by Team Fit)

Rank	PS ID	Title	Org	Fit	Crowd	Feasibility	Verdict
 1	SIH26047	AYUSH Case-Taking Software	AYUSH	95%	200-300	★★★★★	✔ SUBMIT FIRST
 2	SIH26133	Rural Healthcare Access	Maharashtra	85%	200-300	★★★★☆	✔ SUBMIT SECOND
 3	SIH26111	AI Feed & Silage Testing	Dairy	85%	150-200	★★★★☆	✔ SUBMIT THIRD
4	SIH26095	Monitoring & Inspection App	MoSJE	82%	100-180	★★★★★	✔ EASIEST TO BUILD
5	SIH26068	WeatherGPT (Weather AI Chatbot)	IMD	80%	150-250	★★★★☆	✔ STRONG PICK
6	SIH26018	Land Record Digitization	Rural Dev	80%	120-180	★★★★☆	✔ SUBMIT
7	SIH26127	ANPR Traffic Analytics	MoHUA	70%	150-200	★★★★☆	⚠️ AMBITIOUS
8	SIH26106	Email Threat Detection & Forensics	AICTE	65%	200-350	★★★★☆	⚠️ AMBITIOUS
9	SIH26152	Social Media Analytics	NTRO	55%	150-250	★★★★☆	⚠️ RISKY
10	SIH26001	Landslide Early Warning (NER)	MDoNER	45%	120-200	★★★★☆	✗ GIS GAP
11	SIH26208	Student Innovation (Toys/Games)	AICTE	35%	200-400	★★★★☆	✗ TOO VAGUE
12	SIH26014	Land Stack GIS Platform	Rural Dev	35%	100-150	★★★★☆	✗ GIS GAP
13	SIH26013	Geospatial Data Harmonization	Rural Dev	30%	80-120	★★★★☆	✗ GIS GAP

Hardware PS (Ranked by Team Fit)

Rank	PS ID	Title	Org	Fit	Crowd	Verdict
1	SIH26109	Bovine Mastitis IoT+AI	Dairy	60%	40-80	⚠️ Best HW pick
2	SIH26048	iKwath Smart Maker	AYUSH	55%	30-60	⚠️ Viable with prep
3	SIH26008	Conveyor Belt Mining IoT	Steel/NMDC	25%	20-40	❌ AVOID
4	SIH26110	Milk Chilling Can	Dairy	40%	30-50	❌ SKIP (ME only)

Who Suggested What

PS	Analyst	Abhiuday	Om	Arun	Consensus
SIH26047	✓				Analyst Pick
SIH26133		✓			Team Pick (STRONG)
SIH26111	✓				Analyst Pick
SIH26095		✓			Team Pick (STRONG)
SIH26068		✓			Team Pick (STRONG)
SIH26018	✓				Analyst Pick
SIH26106		✓		✓	DOUBLE NOMINATED
SIH26127	✓				Analyst Pick
SIH26014			✓		Om's Pick (AVOID)
SIH26013			✓		Om's Pick (AVOID)
SIH26001				✓	Arun's Pick (MODERATE)
SIH26152		✓			Team Pick (RISKY)
SIH26208		✓			Team Pick (AVOID)
SIH26008		✓			Team Pick (AVOID)

8. Cross-Reference: Which PS Overlap With Past Winners

Based on analysis of 72 SIH winners from 2020-2025:

PS	Similar Past Winner Themes	Pattern Match
SIH26047 (AYUSH)	SIH 2024 Healthcare dashboards, SIH 2023 Patient management	HIGH — Healthcare AI wins consistently
SIH26133 (Healthcare)	SIH 2025 Rural health, SIH 2022 Telemedicine	HIGH — Rural healthcare is perennial winner
SIH26068 (WeatherGPT)	SIH 2024 Disaster management chatbots	MEDIUM — Chatbot PS win when well-executed
SIH26095 (Monitoring)	SIH 2023 Gov monitoring dashboards	MEDIUM — Governance PS win with working demos
SIH26106 (Email Security)	SIH 2025 Crypto detection (winner), SIH 2023 Phishing	HIGH — Cybersecurity PS with AI win big
SIH26018 (Land Records)	SIH 2024 Land digitization	MEDIUM — Government priority area
SIH26111 (Feed Testing)	SIH 2024 AgriTech AI	MEDIUM — Agriculture AI is growing

9. Final Consolidated Recommendation

Priority Order for Idea Submission (Max 5 PS per team)

Priority	PS	Title	Fit	Why
#1	SIH26047	AYUSH Case-Taking Software	95%	4 overlapping projects. Best team-PS alignment in all 226 PS.
#2	SIH26133	Rural Healthcare Access (Maharashtra)	85%	Very similar to SIH26047 — prepare for both simultaneously!
#3	SIH26095	Smart Monitoring & Inspection App	82%	Easiest to build. CRUD + dashboard + mobile = team's bread and butter.
#4	SIH26068	WeatherGPT (Conversational AI)	80%	Om's ARIA AI assistant directly transfers. IMD APIs are well-documented.
#5	SIH26106	Email Threat Detection & Forensics	65%	Abhiuday + Arun both nominated. Anshika's phishing extension is a head-start.

Backup: SIH26018 (Land Digitization), SIH26111 (Feed Testing)

If Pursuing Hardware: SIH26109 (Mastitis IoT+AI) > SIH26048 (iKwath)

❌ PS to DEFINITELY AVOID

- **SIH26013, SIH26014** (GIS/Land Stack) — Zero geospatial skills
- **SIH26008** (Conveyor Belt Mining) — Industrial IoT beyond capability
- **SIH26208** (Student Innovation/Toys) — Too vague, can't prepare
- **SIH26152** (Social Media Analytics/NTRO) — Needs NLP/graph depth team lacks
- **All NTRO deep cybersecurity PS** — Beyond team skill level
- **All Railway PS** — Extreme competition (300+) + domain gap
- **SIH26110** (Milk Chilling Can) — Mechanical engineering only

10. 36-Hour Execution Playbooks

Playbook A: SIH26047 (AYUSH Case-Taking) — Primary

Hours	Aditya	Anshika	Om	Abhiuday	Anoop	Arun
0-4	Flask + MongoDB + Auth	React project + components	AYUSH data model research			
4-10	Patient registration + case history API	AYUSH case-taking form UI	Prakriti questionnaire scoring			
10-18	Treatment planning + classical text search	Flutter mobile app	Symptom-dosha NLP mapping	Dashboard charts		
18-26	Integration + ABDM mockup	Polish UI + testing	Model optimization	Search + export	Docs + testing	AYUSH data
26-32	Unani/Siddha modules	Patient portal	Multilingual (Hindi)	Data viz	API testing	Test data
32-36	Bug fixes + demo prep	UI polish	Demo video	Responsive	Final docs	Presentation

Playbook B: SIH26095 (Monitoring App) — Easiest

Hours	Aditya	Anshika	Om	Abhiuday	Anoop	Arun
0-4	Flask + WebSocket + DB	Flutter project setup	Random assignment algorithm			
4-10	CCTV RTSP integration	Mobile inspection form + geo-tag	VC connector + anomaly detection			
10-18	Real-time monitoring backend	Evidence capture + upload	AI attendance analytics	React dashboard		
18-26	Alert system + notifications	Offline support	Model tuning	Dashboard widgets	Testing	Docs
26-32	Performance optimization	Polish + animations	Edge cases	Charts + maps	API testing	MoSJE data
32-36	Bug fixes	Final UI	Demo	Responsive	Final docs	Presentation

Playbook C: SIH26068 (WeatherGPT) — Strong

Hours	Aditya	Anshika	Om	Abhiuday	Anoop	Arun
0-4	FastAPI + IMD API integration	Flutter project setup	LLM prompt engineering + RAG			
4-10	Weather data pipeline + caching	Chat UI + map	Weather query understanding			
10-18	Alert system + WebSocket push	Voice interaction	Location-based forecasting	Web dashboard		
18-26	Multi-source data aggregation	Offline PWA	Climate trend analysis	D3.js charts	Testing	IMD data
26-32	Multilingual backend	Hindi voice	Agricultural advisories	Responsive	API testing	Use cases
32-36	Bug fixes	Final UI	Demo video	Polish	Docs	Presentation

11. Pre-SIH Preparation Roadmap

Week 1-2: Skill Building

Member	Task
Aditya	Build Flask + MongoDB + React boilerplate. Practice FastAPI. Set up WebSocket demo.
Anshika	Build Flutter app template with offline support (PWA). Practice geo-tagging.
Om	Build a mini WeatherGPT or healthcare chatbot using OpenAI/Gemini API. Practice RAG.
Abhiuday	Build complex React dashboard with charts, tables, WebSocket updates.
Anoop	Learn Postman API testing. Build documentation templates. Practice Git workflow.
Arun	Compile AYUSH data dictionaries. Research Maharashtra PHC structure. Download IMD weather data samples.

Week 3: Domain Research

- Call an AYUSH practitioner (for SIH26047 domain insights)
- Visit a local PHC or health center (for SIH26133 understanding)
- Explore IMD APIs and weather data formats (for SIH26068)
- Research MoSJE schemes and NGO monitoring (for SIH26095)

Week 4: Mock Hackathon

- **12-hour mock build** — build mini version of SIH26047 (patient registration + case history + dashboard)
- Test team dynamics under time pressure
- Identify bottlenecks and prepare pre-built modules
- Practice presentation and demo

Pre-Built Modules to Prepare

- Flask/FastAPI boilerplate with auth, CORS, error handling
- React component library (forms, tables, charts, maps)
- Flutter app shell with navigation, offline storage, camera
- LLM integration template (OpenAI/Gemini with RAG)
- WebSocket real-time update template
- Geo-tagging module (camera + GPS + upload)

12. Appendix: References & Sources

Official Portals

- SIH 2026 Portal: sih.gov.in (<https://sih.gov.in>).
- SIH 2026 All 226 PS PDF: [sih-2026-problem-statements.shaikrohit187.workers.dev](https://sih-2026-problem-statements.shaikrohit187.workers.dev/public/pdfs/SIH_2026_All_PS.pdf) (https://sih-2026-problem-statements.shaikrohit187.workers.dev/public/pdfs/SIH_2026_All_PS.pdf).
- SIH 2026 PS Explorer: csjmu-sih-internal-hackathon.vercel.app (<https://csjmu-sih-internal-hackathon.vercel.app/>).
- SIH 2026 GitHub Dataset: github.com/vedantchalke36/sih-2026-problem-statements (<https://github.com/vedantchalke36/sih-2026-problem-statements>).

Research References

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- AYUSH GCP Guidelines (2013) — CDSCO
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- Bhu-Aadhaar/ULPIN Implementation — 26 states adopted, OpenPlot Blog (2025)
- AI in Indian Railways — JETIR (2024), NASSCOM (2026)
- Dairy Tech ROI — The Bullvine (2025): 91% ROI with AI health sensors
- Smart Urban Infrastructure Market — Future Market Insights (2025-2035)
- NTRO Technical Intelligence — Raksha Anirveda (2024)
- Dark Web Forensics — cyberdeepakyadav.com (2026)

SIH Winner Analysis Sources

- SIH 2024 Grand Finale Results: sih.gov.in/sih2024/sih2024-grand-finale-result (<https://sih.gov.in/sih2024/sih2024-grand-finale-result>).
- SIH 2025 Winners: dev.to/heisdinesh (<https://dev.to/heisdinesh/smart-india-hackathon-2023-winners-471>).
- SIH Winning PPT: slideshare.net (<https://slideshare.net/slideshow/906186982-SIH-2024-Winning-PPT-pptx>).
- SIH 2024 PS Dataset: github.com/Wraient/SIH-2024-Problems (<https://github.com/Wraient/SIH-2024-Problems>).
- How We Won SIH 2022: medium.com/@arinjay_11020 (https://medium.com/@arinjay_11020).

Team GitHub Profiles Analyzed

- Aditya: github.com/Byte-Craftsman-Alpha (<https://github.com/Byte-Craftsman-Alpha>).
- Anshika: github.com/Anshika9838 (<https://github.com/Anshika9838>).
- Om: github.com/tripcoded (<https://github.com/tripcoded>).
- Abhiuday: github.com/Abhiuday02 (<https://github.com/Abhiuday02>).

- Anoop: github.com/anoopshukla01 (<https://github.com/anoopshukla01>)
 - Arun: github.com/arunkumar562816-source (<https://github.com/arunkumar562816-source>)
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Report compiled for Team Paradox | IET Gorakhpur | SIH 2026

18 Problem Statements Analyzed • 6 Team Members Assessed • 3 Research Streams Consolidated

Generated August 27, 2026